

regulatory bodies like FDA/CE. Most device vendors do not recommend their devices for medical usage to avoid getting sued in case of any failure.

Classifying wearables

Wearable devices can be classified into the following four major categories:

Sports and fitness. Sports and fitness devices primarily track movement (using accelerometers) and, in some cases, heartbeat. While worn, these normally track motion and not body parameters.

Personal health monitoring.

These devices track heartbeat, body heat, blood oxygen (SpO2) and ECG (single or 3-lead) signals and, in some cases, have special sensors. These devices normally track body parameters and therefore need to be highly reliable, as readings may be used for lifestyle adjustments.

Tracking and monitoring.

These devices have GSM/GPS functions and are primarily used for tracking the wearer, mostly in geo-fencing applications.

Medical applications. These are high-end devices that are typically worn by patients who need controlled dosages, periodically. These devices are regulated and controlled by regulatory agencies. For this, manufacturers need to follow strict processes during development and manufacture. Importance of the rigid process is to ensure that these devices are traceable and, in case of problem, the root cause can be analysed up to the component level.

Most wearables can handle very low analogue signals and therefore have substantial analogue circuitry to process these. These are also real-time devices and synchronise their operations to the time or date for data collection/measurement. Their functionality can be represented by the flowchart shown in Fig. 2.

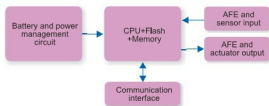


Fig. 1: Parts of a wearable device



Fig. 2: Flowchart showing the functionality of wearable devices

Essential design elements

Let us now take a look at some generic design elements that impact a product. These impact both design and manufacture of wearable devices. A product's performance, its manufacturing and usage characteristics all depend on how well the design is done.

First, let us understand the life cycle of a wearable product so that it is easy to understand the issues involved and how these can be addressed. Fig. 3 shows the typical life cycle of a wearable device. A device essentially goes through four major phases as shown in the figure. While the design, manufacture and use phases are sequential, support phase is basically concurrent, addressing the manufacture and use phases. The reason for taking a life-cycle based approach is to show how designs need to address issues that crop up in subsequent stages.

Design phase. During this phase there are a few things that designers need to know for a successful wearable design. These are:

1. Almost all wearable devices have sensors that use a very low signal level. This necessitates that the designers be very strong in analogue and mixed signal (combination of analogue and digital signals) circuit designs.

2. Since most wearables are battery-operated, designers need to be conversant in low-voltage analogue designs as well as power-conversion techniques (like boost regulators).

3. Intricate knowledge of MCUs used and how their power management is implemented is essential. To reiterate, an increase in processor speed increases power consumption, so managing the processor speed dynamically is crucial for optimal consumption of power.

4. In-depth understanding of the battery types used, their charging characteristics, their behaviour against temperature variation, safety limits and different charging circuits is a must. It is important for designers to know circuit simulation tools like Simulation Program with Integrated Circuit Emphasis (SPICE), as with these tools, they can evaluate the circuitry before committing to the design.

One important thing that a designer needs to keep in mind is the accuracy of the models used in simulation. A wrong or an inadequate model will produce wrong results.

5. With most wearable devices being small in size, the impact of components on neighbouring components in the PCB is very critical. A good example would be the vibrators used in a wearable device that produce strong physical vibrations and also have a magnetic field. Any device nearby that is sensitive to stray magnetic fields can produce wrong results when the vibrator is working.

Designers tend to miss issues like these and struggle to solve them when the actual product runs into trouble. So it is important they understand the components used along with their characteristics.

6. With miniature designs come miniature components and their own specific challenges. Compo-